LAB GUIDELINES

Lab 27 — Disaster Recovery: Replication and Site Failover

Maps to CompTIA Server+ objective **3.8 — Explain the importance of disaster recovery** (site types: hot, cold, warm, cloud, separate geographic locations; replication: constant, background, synchronous vs. asynchronous, application-consistent, file-locking, mirroring, bidirectional; testing: tabletops, live failover, simulated failover, production vs. non-production).

Run all commands on https://killercoda.com/playgrounds/scenario/ubuntu

Required software (free, apt): `rsync`, `lsyncd`, `keepalived`, `inotify-tools`

```
apt update && apt install -y rsync lsyncd keepalived inotify-tools
```

Step 1 — Site types comparison

Site type	Hardware at site	Data at site	Time to bring up	Cost
Hot	Running 24/7	Replicated near-real-time	Minutes	High
Warm	Provisioned, off	Restored from recent backup	Hours	Medium
Cold	Space + power only	Restored from off-site tape	Days	Low
Cloud	Provisioned on-demand	Replicated to cloud	Minutes to hours	Variable

DR sites must be in **separate geographic locations** — far enough that one disaster (flood, fire, regional outage) does not affect both.

Step 2 — Asynchronous replication with rsync (background)

Simulate two sites with two directories:

```
mkdir -p /srv/primary /srv/dr
echo "v1" > /srv/primary/app.txt

# Background async replication every 60 s via cron / lsyncd
rsync -avh --delete /srv/primary/ /srv/dr/
ls /srv/dr/
```

Asynchronous = primary acknowledges the write before the replica has it. Cheaper, but the **RPO** = the replication lag.

Step 3 — Near-synchronous replication with `lsyncd`

`lsyncd` watches the source with inotify and pushes deltas to the target within seconds:

```
cat > /etc/lsyncd/lsyncd.conf.lua <<'EOF'
settings { logfile = "/var/log/lsyncd.log", statusFile = "/var/run/lsyncd.status" }
sync {
    default.rsync,
    source = "/srv/primary",
    target = "/srv/dr",
    delay  = 2
}
EOF
systemctl restart lsyncd
echo "v2" > /srv/primary/app.txt
```

```
sleep 4
cat /srv/dr/app.txt        # should print v2
```

Step 4 — Synchronous replication concept

Synchronous replication blocks the write until **both** sites confirm — RPO = 0. Used for:

- **DRBD** (block-level replicated devices — Linux)
- Storage-array sync replication (NetApp SnapMirror sync, Dell SRDF)
- **MySQL semi-sync**, **PostgreSQL synchronous_commit = remote_apply**

Synchronous is **distance-bounded**: latency adds to every write. Practical limit is ~ 50–100 km.

Step 5 — Constant vs. background

Mode	What it means
Constant	Replication is always running — every change ships ASAP
Background	Periodic batches (e.g. 5-min rsync) — lower load, higher RPO

Constant ≈ what `lsyncd` does. Background ≈ cron-driven rsync.

Step 6 — Application-consistent replication and file locking

Server+ 3.8 lists **application consistent**, **file locking**, **mirroring**, **bidirectional**.

- **Application-consistent** = replica is the database in a state the DB will recover cleanly (flushed logs, no in-flight transactions). Done by quiescing the app, snapshotting, then shipping the snapshot.
- **File locking** = replication tool must respect locks so it does not capture half-written files.
- **Mirroring** = one-way primary → replica.
- **Bidirectional** = both sites can accept writes. Powerful but introduces **conflict resolution** that the storage / app must handle.

Step 7 — VIP failover with keepalived (DR cut-over plumbing)

On a real DR pair, the application VIP floats. With `keepalived` (see Lab 14):

```
PRIMARY  state MASTER  priority 150  → owns VIP 10.99.99.99
DR       state BACKUP  priority 100  → takes VIP if heartbeat fails
```

A failure of the master triggers BACKUP to **gratuitous-ARP** the VIP and take over within seconds. The application reconnects to the same VIP, now at the DR site.

Step 8 — DR testing — Server+ lists four types

Test	What happens	Risk
Tabletop	Team walks through the runbook in a meeting room	None
Simulated failover	Failover into an isolated test environment	Low
Live failover	Production traffic is moved to the DR site	High
Production vs. non-production	Decide which environment you'll test in — never your first live failover during a real outage	—

Schedule: tabletop quarterly, simulated yearly, live every 1–2 years (or as audit demands).

Step 9 — Tabletop DR exercise script (run this with the team)

```
SCENARIO: Primary data centre has a building-wide power outage estimated 12 hours.
INJECTS:
  T+00m  Power lost. Monitoring alerts fire.
  T+05m  Confirm with facilities — outage real, ETA 12h.
  T+15m  Declare disaster. Open bridge call.
  T+30m  Begin DR runbook step 1: redirect DNS to DR site.
  T+45m  Promote DR DB replica to primary (app-consistent? yes/no — discuss).
  T+60m  Service back up at DR. Confirm RTO target.
  T+12h  Power restored. Begin fail-back: replicate writes back to primary, then cut over.
DECISIONS TO MAKE:
  - Who declares the disaster?
  - When do we communicate to customers?
  - When is the SLA breached?
```

Document every gap revealed in the exercise — those are the **action items**.

Step 10 — RPO / RTO vs. replication choice

Replication strategy	Achievable RPO	Cost
Tape off-site weekly	1 week	Low
Nightly rsync to cloud	24 hours	Low
`lsyncd` near-real-time	~10 seconds	Medium
Sync block-level (DRBD/SRDF)	0	High

Pick the cheapest one that meets the **RPO from the BIA** (Lab 17). Buying more than required is wasted spend.

Free online tools

- **NIST SP 800-34 (Contingency Planning)** — https://csrc.nist.gov/publications/detail/sp/800-34/rev-1/final
- **DRII Professional Practices** — https://drii.org/
- **lsyncd manual** — https://axkibe.github.io/lsyncd/
- **DRBD docs** — https://docs.linbit.com/

What you learned

- Hot / warm / cold / cloud site comparison with cost and time-to-recover.
- Async (rsync, lsyncd) vs. sync (DRBD, SRDF, semi-sync DB) replication trade-offs.
- Application-consistent, file-locking, mirroring, bidirectional replication semantics.
- VIP failover plumbing with keepalived (cut-over mechanics).
- Four DR test types and a ready-to-run tabletop script.
- Matching the replication strategy to the BIA-driven RPO/RTO numbers.